

Sentinel PX-200 Pump Controller

Synthetic demonstration manual | Version 1.0 | Model PX-200

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1. Safety prerequisites

Qualified personnel only. Before opening the enclosure, stop the pump through the normal control sequence, isolate the upstream electrical disconnect, apply the site's lockout/tagout procedure, and verify absence of voltage with an approved tester. Wear the PPE required by the site risk assessment.

Never bypass the door interlock, thermal switch, or motor-protection relay. Do not restart equipment when the cause of an over-temperature event remains unknown.

2. Status and fault codes

Code	Meaning	Immediate response
E-11	Inlet pressure below configured minimum	Stop the pump and inspect the supply path for a closed valve, blockage, or empty source tank.
E-17	Controller cooling airflow restricted	Allow the controller to cool. After isolation, inspect the enclosure filter, fan intake, cooling fan, and J4.
E-23	Motor current imbalance	Stop operation; inspect motor leads, terminals, and supply phase balance.
W-05	Maintenance interval approaching	Schedule the applicable preventive-maintenance work order.

3. E-17 troubleshooting procedure

E-17 is raised when heat-sink temperature exceeds **70 C for more than 30 seconds** while the cooling-fan command is active.

1. Stop the pump using the normal control sequence.
2. **Isolate electrical energy, apply lockout/tagout, and verify absence of voltage** before opening the enclosure.
3. Wait at least five minutes for internal components to cool.
4. Inspect the air inlet and enclosure filter. Clean or replace a blocked filter.
5. Confirm the fan rotor turns freely and no cable or debris obstructs it.
6. Inspect cooling-fan connector **J4** for looseness, corrosion, or damaged conductors.
7. Restore the enclosure, remove lockout/tagout under site procedure, and run the diagnostic fan test.
8. Return to service only when the fan test passes and heat-sink temperature remains below **55 C for ten minutes**.

If E-17 returns, stop the controller and escalate to an authorized service technician. Do not repeatedly reset the fault.

4. Normal-condition cooling inspection

Under normal indoor conditions, inspect the enclosure filter every **500 operating hours**. Nominal fan speed is 2,400-3,200 rpm. A commanded speed below 1,800 rpm for more than 20 seconds triggers the fan diagnostic warning.

5. Connector reference

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Connector	Function	Inspection notes
J2	Inlet pressure sensor	Confirm the shield is terminated at the controller end only.
J4	Cooling fan power and tachometer	Check latch engagement, pin seating, corrosion, and conductor damage.
J7	Motor current sensors	Do not disconnect while energized.

Diagnostic interpretation

A fan command with no tachometer feedback can be caused by a loose J4 connection, damaged conductors, rotor obstruction, or a fan fault. Do not replace the fan solely from an E-17 indication; complete the procedure and diagnostic fan test.

6. Simplified cooling path

Keep at least **150 mm** clearance around the inlet and exhaust openings.

**FILTERED LOWER INLET → J4-POWERED FAN → HEAT
SINK → UPPER EXHAUST**

Figure 1: Cooling path from the filtered lower inlet through the J4-powered fan and heat sink to the upper exhaust. This caption is indexed by ProofRAG; it is not full diagram reasoning.

Inspection notes

Check that inlet and exhaust openings remain unobstructed. Inspect the filter media, fan rotor, and nearby cabling only after the safety prerequisites on page 1 are satisfied.